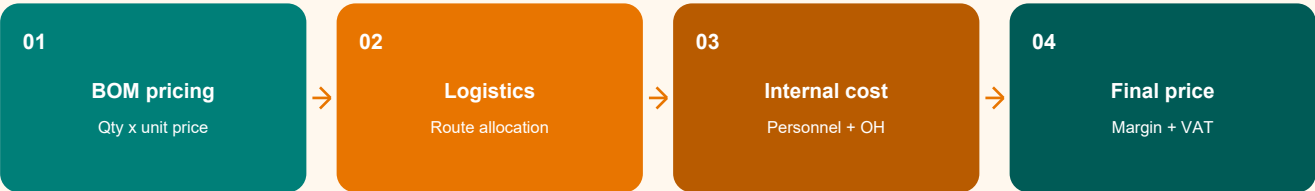


# LUMA BOM Manager

## Price Calculation Executive Summary

### Purpose

This document explains the commercial calculation path from the active project BOM to the final customer price. It is limited to the executive logic, formulas, validation gates, and calculation order.



### Executive outcome

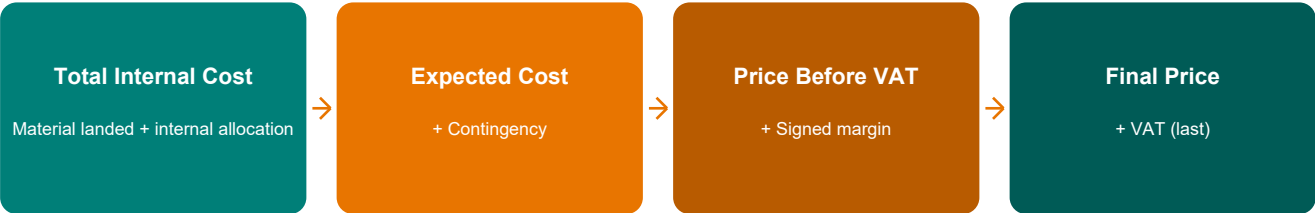
The final result is traceable: every price starts from a BOM quantity, an approved supplier unit price, an approved logistics route, or an administrator-maintained internal cost. Missing commercial data blocks the final price and is never silently interpreted as zero.

| Pricing layer  | Primary source                                              | Result                      |
|----------------|-------------------------------------------------------------|-----------------------------|
| Material       | Active Project BOM + selected active supplier Price List    | Section material cost       |
| Transportation | Selected active logistics rate + project weight or quantity | Allocated project logistics |
| Internal       | Personnel and Overhead settings + project MWp               | Project internal allocation |
| Commercial     | Contingency, signed Margin, and VAT selection               | Final customer price        |

Prepared 01 September 2026  
Currency basis for the current Result: EUR

# 1. Executive summary

LUMA BOM Manager builds the final price in controlled layers. Material starts with the engineering BOM but uses commercial supplier data for value. Logistics is added by shipment route. Personnel and Overhead are allocated using the active project capacity. Contingency and Margin are then applied in sequence, and VAT is added once as the last calculation.



## Commercial package structure

| Result section  | Included commercial areas                                     |
|-----------------|---------------------------------------------------------------|
| Steel Structure | Posts + Substructure                                          |
| Bearing         | Bearing                                                       |
| Slew Drive      | Slew Drive                                                    |
| Fasteners       | Fasteners                                                     |
| Electrical      | Electrical + PV Module + Limit Switch + SOLTRK + Junction Box |

## Key control principles

- + Supabase commercial records remain the source of supplier prices, logistics rates, Personnel cost, and Overhead cost.
- + The active project supplies BOM quantities, project MWp, selected suppliers and routes, Contingency, signed Margin, and VAT.
- + The Result currently combines EUR only. A non-EUR price or logistics rate blocks the final result until an approved exchange-rate method exists.
- + A missing supplier, active Price List, unit price, or required route blocks completion. Missing values are not converted to zero.

### Decision rule

The calculation may display partial subtotals for review, but the final price is valid only when every required input layer is complete.

## 2. Material and logistics calculation

1

### Read the active project BOM

Use the engineering result as the quantity authority.

**Required quantity(i) = active Project BOM quantity for item(i)**

- + Only items with a required quantity greater than zero enter commercial pricing.
- + Supplier name is not used as a permanent item identifier; BOM TAGs connect items to Price List records.

2

### Price each BOM line

Apply the selected active supplier Price List to every required TAG.

**Line cost(i) = BOM quantity(i) x supplier unit price(i)**

- + Section material cost = sum of all line costs in that section.
- + A blank unit price is missing data, not zero. One missing required price makes the section incomplete.
- + The Price List and its supplier/category assignment must be active and valid for selection.

3

### Calculate shipment demand

Convert the project requirement into a route capacity basis.

**Containers = ceiling(project shipment amount / container capacity)**

- + Steel Structure uses BOM weight in kg.
- + Bearing and Slew Drive use authoritative BOM quantities in pcs.
- + The displayed full-container cost uses the rounded container count; the Result uses allocated project logistics based on actual capacity utilization.

4

### Build logistics cost per container

Add the route segments, handling, storage, customs, and insurance configured for the selected route.

**Total/container = route transport + warehouse handling + customs + insurance**

- + Warehouse handling may include port-to-warehouse transport, unloading, storage periods, truck loading, and warehouse-to-site transport.
- + Storage periods = ceiling(storage days / storage period days).
- + Allocated logistics = (Total/container / capacity) x actual project shipment amount.

### 3. Landed material and internal allocation

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#### Allocate transportation to Result columns

Keep the route build-up visible instead of hiding all transportation in one number.

**Section landed cost = material cost + Genoa + Italian/Warehouse + FOB China**

- + FOB China includes origin-to-port and FOB-related customs/insurance allocation.
- + Genoa to Site includes port-to-site transport and the matching direct route allocation.
- + Italian Supplier/Warehouse to Site includes warehouse routing, unloading, storage, reloading, and final site delivery.
- + Required logistics currently applies to Posts, Substructure, Bearing, and Slew Drive. Other section transport remains zero unless configured in a future phase.

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#### Calculate material totals and shares

Show both the material mix and each section's contribution to all internal cost.

**Material landed total = sum(section landed cost)**

- + Share in Material (%) = section landed cost / material landed total x 100.
- + Share in Total (%) = section landed cost / total internal cost x 100.

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#### Allocate Personnel cost

Convert administrator-maintained monthly costs into a project amount through the yearly target and project capacity.

**Personnel project cost = (monthly cost x 12 / yearly target MW) x project MWp**

- + Applied rows: Engineering, Test/Commissioning, Project Management, and Admin Management.
- + The Finance row is not included.
- + Personnel Cost per MW = yearly Personnel cost / yearly target MW.

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#### Allocate Overhead cost

Spread annual company Overhead across annual project capacity, then apply it to the active project.

**Overhead project cost = (annual Overhead / annual capacity MWp) x project MWp**

- + Overhead yearly total is the sum of the active Overhead register values.
- + Overhead Cost per MW is read from the same formula in Analysis and Result.

## 4. Final price sequence

### Mandatory order

Contingency is applied to Total Internal Cost. Margin is applied to Expected Cost and may be negative or positive. VAT is applied once, after Margin, to Price Before VAT.

| Step | Calculation         | Formula                                                                 |
|------|---------------------|-------------------------------------------------------------------------|
| 9    | Total Internal Cost | Material landed total + Personnel project total + Overhead project cost |
| 10   | Contingency Amount  | Total Internal Cost x Contingency %                                     |
| 11   | Expected Cost       | Total Internal Cost + Contingency Amount                                |
| 12   | Margin Amount       | Expected Cost x signed Margin %                                         |
| 13   | Price Before VAT    | Expected Cost + Margin Amount                                           |
| 14   | VAT Amount          | Price Before VAT x selected VAT rate (0%, 10%, or 22%)                  |
| 15   | Final Price         | Price Before VAT + VAT Amount                                           |

### Compact formula

#### Final Price

$[\text{Total Internal Cost} \times (1 + \text{Contingency \%})] \times (1 + \text{signed Margin \%}) \times (1 + \text{VAT rate})$

### Readiness checklist before accepting the result

| Check             | Required condition                                                                                           |
|-------------------|--------------------------------------------------------------------------------------------------------------|
| BOM               | Active project calculation is complete and required quantities are available.                                |
| Supplier pricing  | Every required section has a selected eligible supplier and active Price List; every required TAG is priced. |
| Currency          | All material and logistics values used by Result are EUR.                                                    |
| Logistics         | Every required shipment has an active complete route and valid capacity data.                                |
| Personnel         | Monthly costs, yearly target MW, and project MWp are valid.                                                  |
| Overhead          | Annual Overhead and annual project capacity are valid.                                                       |
| Commercial inputs | Contingency is zero or positive; Margin is numeric; VAT is 0%, 10%, or 22%.                                  |

#### Final control

If any readiness check fails, the application should show the reason and keep Final Price incomplete. The correct response to missing data is correction or approval - never an invented zero or exchange rate.